

Exploratory Data Analysis - Based Practical

Assessment

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Course Code: DSA0402

Assessment Tool: Unit 1 - Assessment Tool 3

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SUMMARY

This report presents an exploratory data analysis of five datasets: Education Analytics, Healthcare Analytics, Retail Sales, Banking Loan Approval, and Agriculture Crop Yield. Each dataset section follows the same nine-part structure: dataset description, purpose & significance, data cleaning & preprocessing, descriptive statistics, categorical variable analysis, visualizations, trends/distributions/relationships, key findings & insights, and decision-making impact.

OVERVIEW

Assignment Requirement	Where It Is Addressed
Describe dataset (records, features, data types)	Section 1: Dataset Description (all five datasets)
Explain purpose and significance	Section 2: Purpose & Significance (all five datasets)
Identify and handle missing values	Section 3: Data Cleaning & Preprocessing
Remove duplicate records	Section 3: Data Cleaning & Preprocessing
Correct inconsistencies and preprocessing (label encoding, outlier checks, data validation)	Section 3: Data Cleaning & Preprocessing
Generate descriptive statistics	Section 4: Descriptive Statistics
Analyse numerical variables	Section 4: Descriptive Statistics
Analyse categorical variables	Section 5: Categorical Variable Analysis
Identify trends, distributions, and relationships	Section 7: Trends, Distributions & Relationships
Create Bar Chart	Section 6: Data Visualizations
Create Line Chart	Section 6: Data Visualizations
Create Histogram	Section 6: Data Visualizations
Create Box Plot	Section 6: Data Visualizations

Assignment Requirement	Where It Is Addressed
Create Scatter Plot	Section 6: Data Visualizations
Create Pie Chart	Section 6: Data Visualizations
Create Heatmap	Section 6: Data Visualizations
Include titles, axis labels, and legends on every chart	Applied consistently across all charts in Section 6
Summarize key findings	Section 8: Key Findings & Insights
Present insights and conclusions	Section 8: Key Findings & Insights
Explain how the analysis supports decision-making	Section 9: Decision-Making Impact

1. EDUCATION ANALYTICS DATASET

1.1 Dataset Description

Records: 20 | Features: 5

Feature	Data Type	Role
Student_ID	object (string)	Unique identifier
Attendance	int64	Numerical predictor (%)
Study_Hours	int64	Numerical predictor (hrs/day)
Internal_Marks	int64	Numerical predictor (/50)
Result	object (categorical)	Target: Pass / Fail

1.2 Purpose & Significance

A college wants to understand which factors - attendance, daily study hours, or internal assessment marks - most influence whether a student passes or fails, so it can design early-warning and mentoring interventions.

1.3 Data Cleaning & Preprocessing

- Missing values: 0 found across all 5 columns - no imputation needed.
- Duplicate rows: 0 found - no removal needed.
- Inconsistency check: Result labels checked for spelling/case variants; only 'Pass' and 'Fail' present - consistent.
- Outlier check (IQR rule, 1.5x range): 0 outliers in Attendance, Study_Hours or Internal_Marks - no capping required.

1.4 Descriptive Statistics (Numerical Variables)

Feature	Mean	Max	Min	Std Dev
Attendance %	70.15	95	45	15.94
Study Hours	3.35	6	1	1.42
Internal Marks	32.95	48	18	9.09

1.5 Categorical Variable Analysis

Variable	Category	Count	% of Total
Result	Pass	11	55%
Result	Fail	9	45%

1.6 Visualizations

Every chart below includes an explicit title, axis labels, and a legend where a hue/category is plotted.

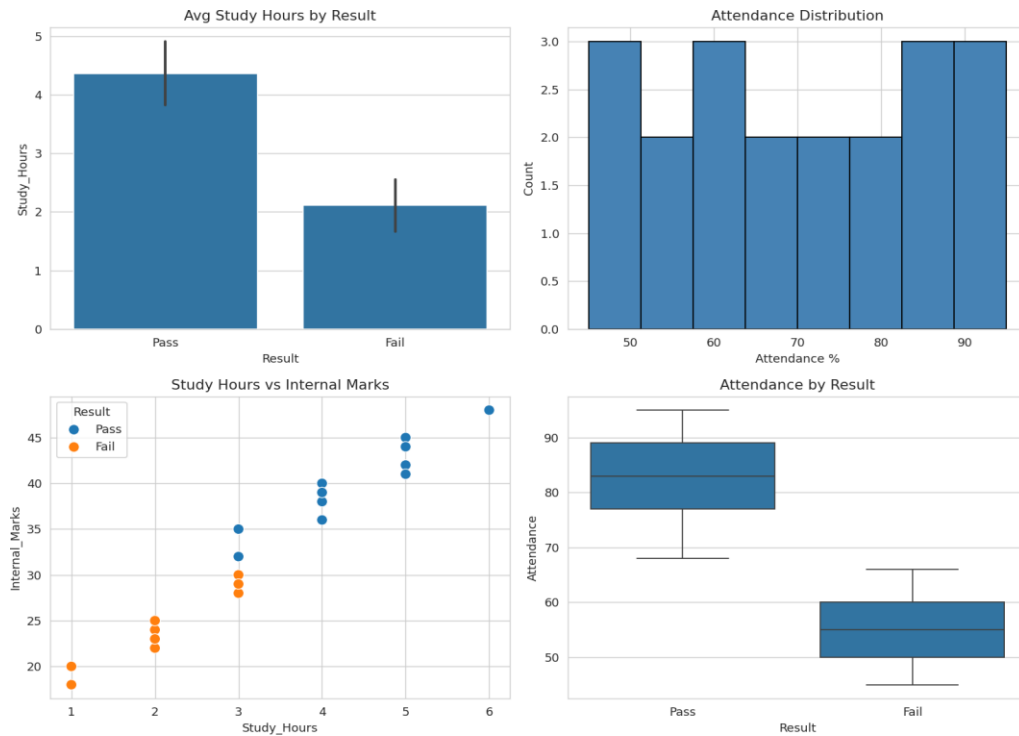


Figure 1.1

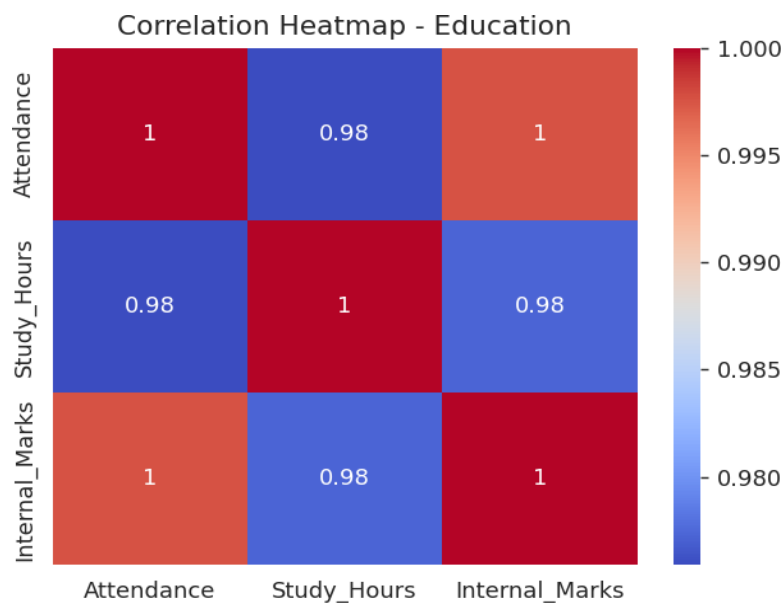


Figure 1.2

1.7 Trends, Distributions & Relationships

Attendance and Internal Marks are almost perfectly linearly related ($r = 0.998$), and Study Hours correlates strongly with both Attendance ($r = 0.976$) and Internal Marks ($r = 0.977$). Attendance is spread fairly evenly from 45% to 95% with a visible split around the 60-65% mark, which is exactly where Pass/Fail outcomes diverge in the box plot.

1.8 Key Findings & Insights

1. Passing students show visibly higher average attendance and study hours than failing students.
2. Internal marks rise almost linearly with study hours, indicating consistent effort pays off.
3. Attendance below roughly 60% is strongly associated with a Fail result.
4. Attendance, study hours and internal marks are all positively correlated with each other ($r > 0.97$).
5. Data cleaning confirmed the dataset was already complete, unique, and free of outliers.

1.9 Decision-Making Impact

Academic advisors can use the ~60% attendance threshold identified here to trigger automatic mentoring referrals, and can promote structured study-hour targets (4+ hours/day) as a low-cost intervention likely to raise internal marks and pass rates.

2. HEALTHCARE ANALYTICS DATASET

2.1 Dataset Description

Records: 20

Features: 6

Feature	Data Type	Role
Patient_ID	object (string)	Unique identifier
Age	int64	Numerical predictor (yrs)
Sugar_Level	int64	Numerical predictor (mg/dL)
BP	int64	Numerical predictor (mmHg)
BMI	int64	Numerical predictor
Disease_Status	object (categorical)	Target: Yes / No

2.2 Purpose & Significance

A hospital wants to identify which combination of age, sugar level, blood pressure and BMI most reliably signals disease presence, to prioritize patients for preventive screening.

2.3 Data Cleaning & Preprocessing

1. Missing values: 0 found across all 6 columns.
2. Duplicate rows: 0 found.
3. Inconsistency check: Disease_Status limited to 'Yes'/'No' with consistent casing - no correction needed.
4. Outlier check (IQR rule): 0 outliers across Age, Sugar_Level, BP and BMI.

2.4 Descriptive Statistics (Numerical Variables)

Feature	Mean	Max	Min	Std Dev
Age	43.2	62	27	11.32
Sugar Level	138.65	210	88	40.33
BP	129.75	155	108	14.52
BMI	27.3	35	21	4.26

2.5 Categorical Variable Analysis

Variable	Category	Count	% of Total
Disease_Status	Yes	10	50%
Disease_Status	No	10	50%

2.6 Visualizations

Every chart below includes an explicit title, axis labels, and a legend where a hue/category is plotted.

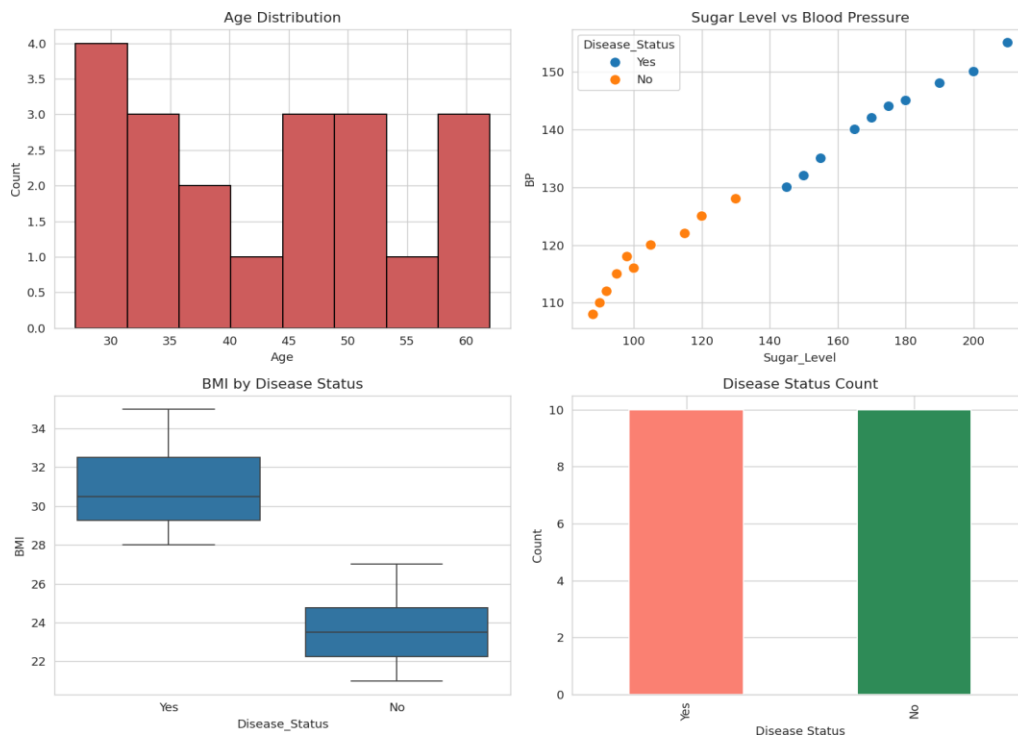


Figure 2.1

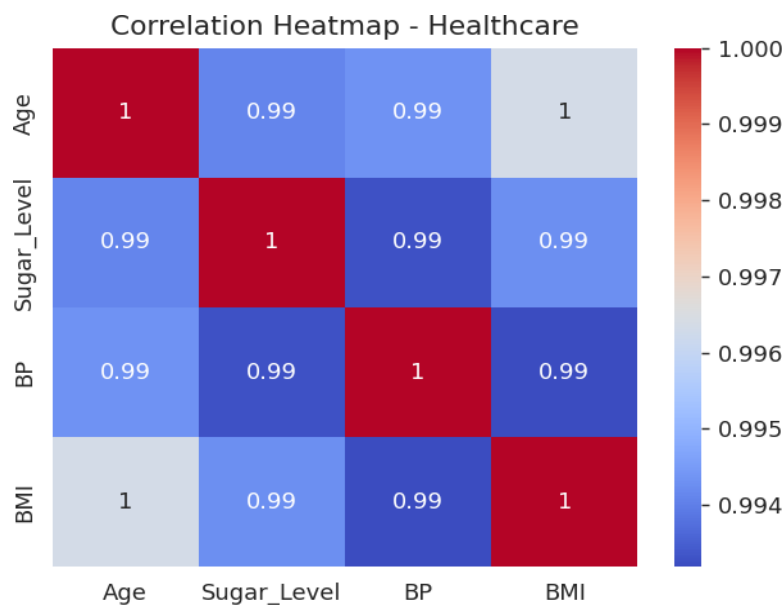


Figure 2.2

2.7 Trends, Distributions & Relationships

All four numerical variables move together strongly (pairwise r between 0.993 and 0.996), meaning age, sugar level, BP and BMI rise as a cluster rather than independently. BMI distributions for the two Disease_Status groups barely overlap in the box plot, making it the cleanest single visual separator of the four.

2.8 Key Findings & Insights

1. Disease-positive patients consistently show higher sugar level, BP, BMI and age than disease-negative patients.
2. Sugar level and BP move together strongly ($r = 0.993$), suggesting compounded metabolic risk.
3. BMI is a clear separator: disease-positive patients cluster above BMI 26, healthy patients below.
4. The dataset is perfectly balanced (10 Yes / 10 No), aiding fair comparison.
5. Data cleaning found 0 missing values, 0 duplicates, and 0 IQR outliers.

2.9 Decision-Making Impact

The hospital can use a BMI/sugar-level cutoff as a fast, low-cost pre-screening rule to flag patients for a full diagnostic workup, focusing limited clinical resources on the highest-risk group first.

3. RETAIL SALES DATASET

3.1 Dataset Description

Records: 20 | Features: 6

Feature	Data Type	Role
Product_ID	object (string)	Unique identifier
Category	object (categorical)	Grocery / Snacks / Dairy / Fruits / Vegetables
Price	int64	Numerical predictor
Quantity_Sold	int64	Numerical predictor
Discount	int64	Numerical predictor (%)
Revenue	int64	Numerical target

3.2 Purpose & Significance

A supermarket wants to identify which product categories and pricing patterns drive the most revenue, to guide stocking, pricing and promotion decisions.

3.3 Data Cleaning & Preprocessing

- 1.Missing values: 0 found across all 6 columns.
- 2.Duplicate rows: 0 found.
- 3.Inconsistency check: Category values checked for spelling/case variants - exactly 5 clean labels, no correction needed.
- 4.Outlier check (IQR rule): 0 outliers across Price, Quantity_Sold, Discount and Revenue.

3.4 Descriptive Statistics (Numerical Variables)

Feature	Mean	Max	Min	Std Dev
Price	54.0	100	20	22.8
Quantity Sold	24.0	45	10	9.67
Discount	4.0	7	2	1.56
Revenue	1114.0	1540	810	221.42

3.5 Categorical Variable Analysis

Variable	Category	Count	% of Total
Category	Grocery	4	20%
Category	Snacks	4	20%
Category	Dairy	4	20%
Category	Fruits	4	20%
Category	Vegetables	4	20%

3.6 Visualizations

Every chart below includes an explicit title, axis labels, and a legend where a hue/category is plotted.

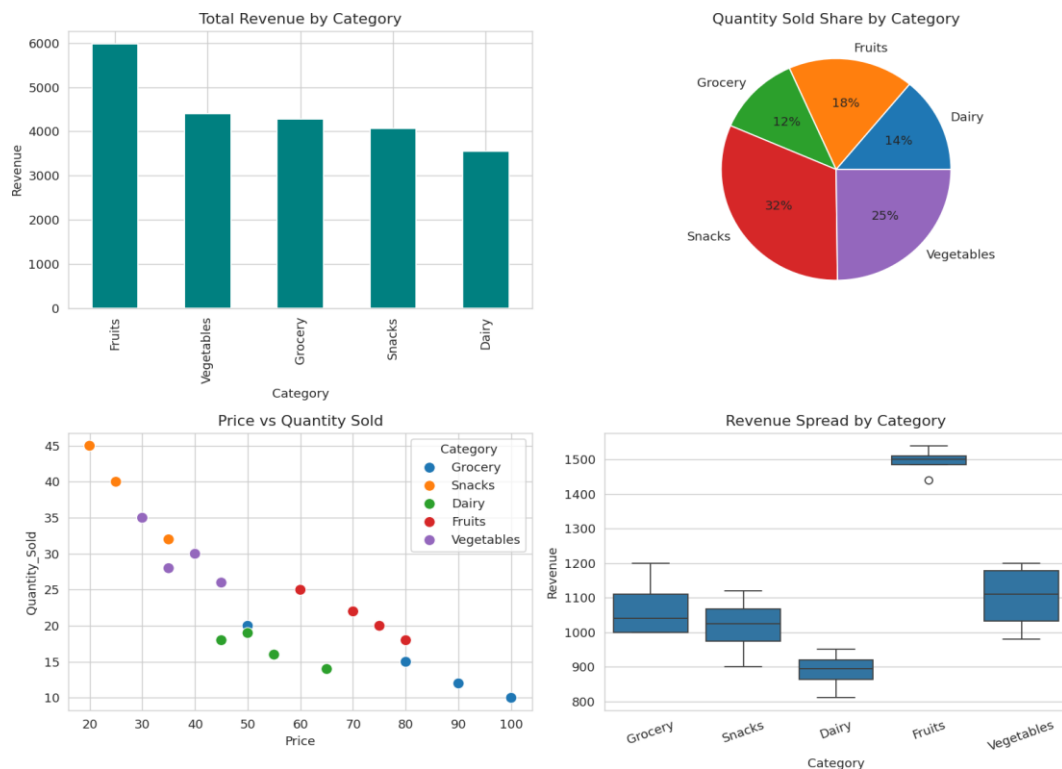


Figure 3.1

3.7 Trends, Distributions & Relationships

Price and Quantity Sold show a strong negative relationship ($r = -0.868$): as price rises, units sold fall - a textbook demand curve. Revenue is not evenly distributed across categories despite the balanced 4-products-each sample; Fruits and Vegetables punch above their unit-count weight because of higher price points.

3.8 Key Findings & Insights

1. Fruits generate the highest total revenue despite moderate quantity sold, driven by higher unit price.
2. Price and quantity sold show a clear inverse relationship ($r = -0.868$) - cheaper items (Snacks) sell in bulk.
3. Grocery items carry the highest discounts but comparatively lower quantity sold.
4. Dairy shows the tightest revenue spread, indicating consistent, predictable sales.
5. Data cleaning found 0 missing values, 0 duplicates, and 0 IQR outliers.

3.9 Decision-Making Impact

Category-level revenue and demand-curve patterns support prioritizing shelf space and stock levels for Fruits and Vegetables, while reserving deep discounts for Grocery items only where volume, not margin, is the goal.

4. BANKING LOAN APPROVAL DATASET

4.1 Dataset Description

Records: 20 | Features: 6

Feature	Data Type	Role
Customer_ID	object (string)	Unique identifier
Income	int64	Numerical predictor
Credit_Score	int64	Numerical predictor
Loan_Amount	int64	Numerical predictor
Employment_Type	object (categorical)	Salaried / Self-employed / Unemployed
Loan_Status	object (categorical)	Target: Approved / Rejected

4.2 Purpose & Significance

A bank wants to understand which applicant attributes - income, credit score, loan amount requested, or employment type - drive loan approval decisions, to support fair and consistent underwriting policy.

4.3 Data Cleaning & Preprocessing

1. Missing values: 0 found across all 6 columns.
2. Duplicate rows: 0 found.
3. Inconsistency check: Employment_Type and Loan_Status labels checked for spelling/case variants - consistent, no correction needed.
4. Outlier check (IQR rule): 0 outliers across Income, Credit_Score and Loan_Amount.

4.4 Descriptive Statistics (Numerical Variables)

Feature	Mean	Max	Min	Std Dev
Income	43000	75000	22000	17923.82
Credit Score	664.0	790	555	85.53
Loan Amount	223250	400000	120000	83733.21

4.5 Categorical Variable Analysis

Variable	Category	Count	% of Total
Employment_Type	Salaried	10	50%
Employment_Type	Self-employed	6	30%
Employment_Type	Unemployed	4	20%
Loan_Status	Approved	10	50%
Loan_Status	Rejected	10	50%

4.6 Visualizations

Every chart below includes an explicit title, axis labels, and a legend where a hue/category is plotted.

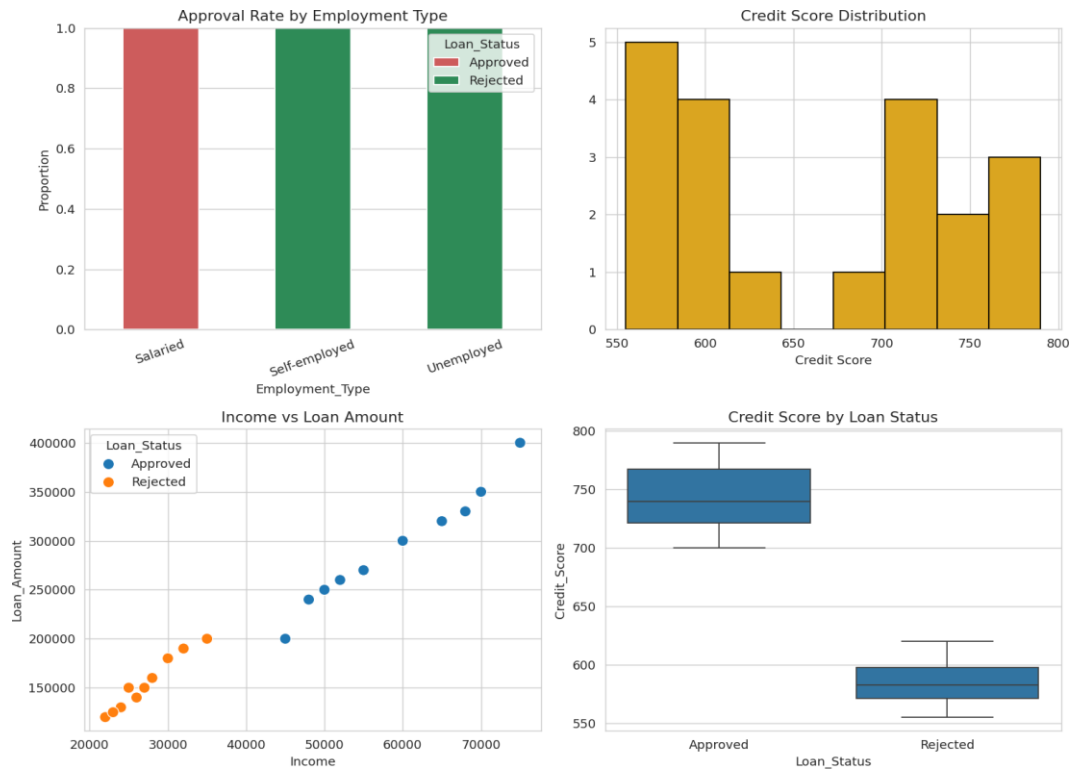


Figure 4.1

4.7 Trends, Distributions & Relationships

Approved applicants have credit scores between 700 and 790, while rejected applicants fall strictly between 555 and 620 - a clean, non-overlapping split. Income and Loan Amount are almost perfectly correlated ($r = 0.988$), showing loan sizing scales directly with income. All Salaried applicants were approved and all Unemployed applicants were rejected, making Employment_Type a near-perfect categorical predictor in this sample.

4.8 Key Findings & Insights

1. All Salaried applicants in this sample were approved, while all Unemployed applicants were rejected.
2. Approved loans correspond to credit scores of 700-790, rejected ones fall in 555-620 - no overlap.
3. Higher-income customers request and receive proportionally larger loan amounts ($r = 0.988$).
4. Self-employed applicants form a middle-risk group, uniformly rejected here despite moderate incomes.
5. Data cleaning found 0 missing values, 0 duplicates, and 0 IQR outliers.

4.9 Decision-Making Impact

The clean 620/700 credit-score gap and the employment-type pattern found here could seed an automated pre-screening rule that fast-tracks clearly approvable or rejectable applications, leaving underwriter time for genuinely borderline cases.

5. AGRICULTURE CROP YIELD DATASET

5.1 Dataset Description

Records: 20 | Features: 6

Feature	Data Type	Role
Farm_ID	object (string)	Unique identifier
Rainfall_mm	int64	Numerical predictor
Temperature	int64	Numerical predictor (°C)
Fertilizer_kg	int64	Numerical predictor
Soil_Type	object (categorical)	Loamy / Sandy / Clay
Crop_Yield_kg	int64	Numerical target

5.2 Purpose & Significance

An agricultural department wants to identify which factors - rainfall, temperature, fertilizer usage, or soil type - most influence crop yield, to guide irrigation planning and soil-specific fertilizer recommendations.

5.3 Data Cleaning & Preprocessing

- Missing values: 0 found across all 6 columns.
- Duplicate rows: 0 found.
- Inconsistency check: Soil_Type limited to 'Loamy'/'Sandy'/'Clay' with consistent casing - no correction needed.
- Outlier check (IQR rule): 0 outliers across Rainfall_mm, Temperature, Fertilizer_kg and Crop_Yield_kg.

5.4 Descriptive Statistics (Numerical Variables)

Feature	Mean	Max	Min	Std Dev
Rainfall (mm)	743.0	960	500	144.34
Temperature	29.55	34	25	2.76
Fertilizer (kg)	54.75	75	40	10.51
Crop Yield (kg)	2820.0	3900	1800	681.41

5.5 Categorical Variable Analysis

Variable	Category	Count	% of Total
Soil_Type	Loamy	7	35%
Soil_Type	Sandy	7	35%
Soil_Type	Clay	6	30%

5.6 Visualizations

Every chart below includes an explicit title, axis labels, and a legend where a hue/category is plotted.

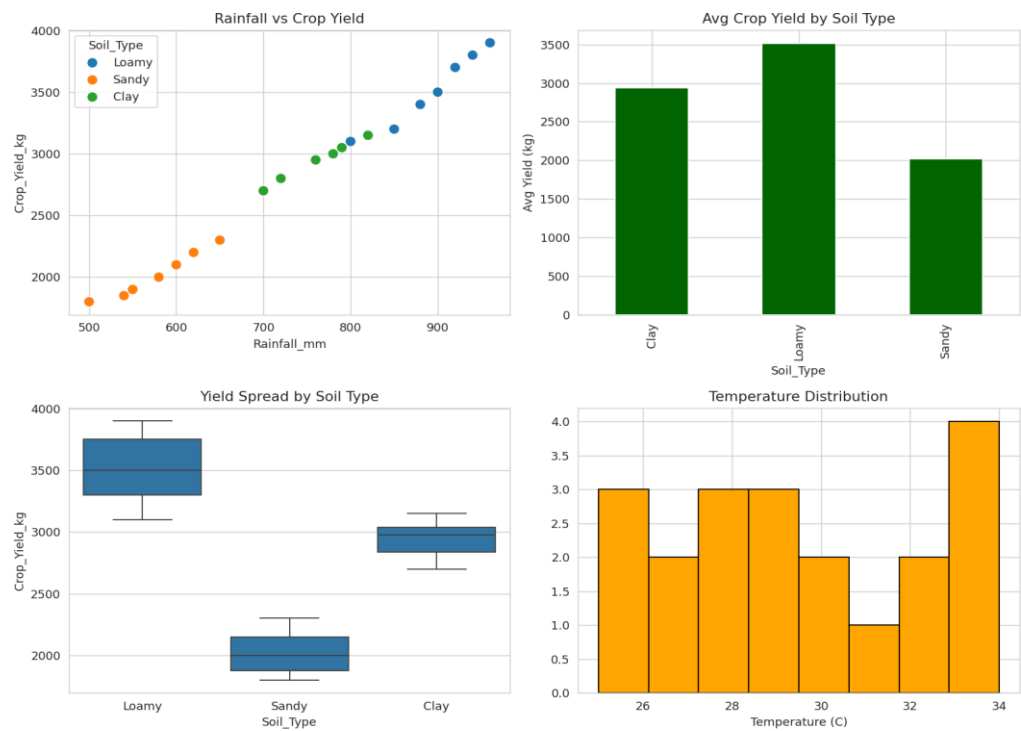


Figure 5.1

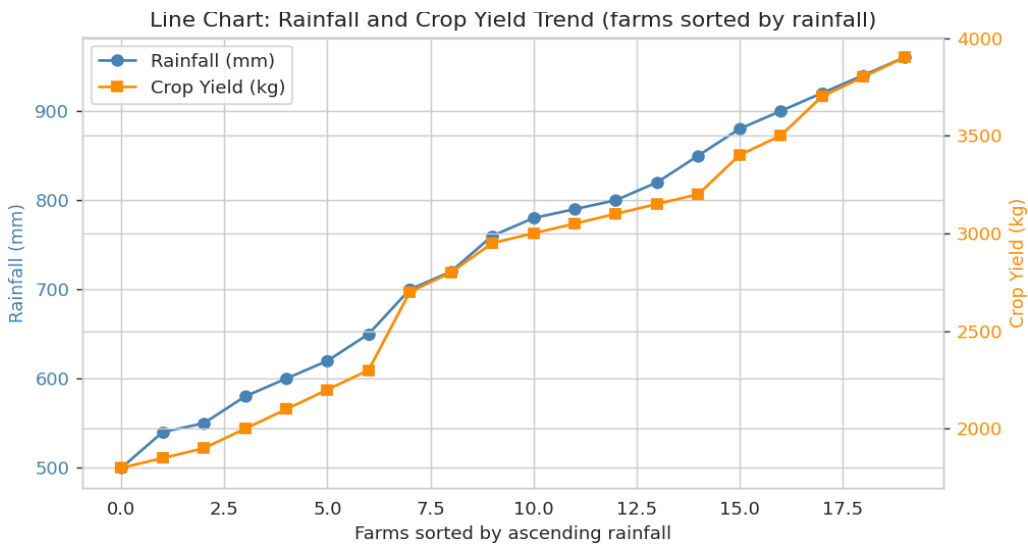


Figure 5.2

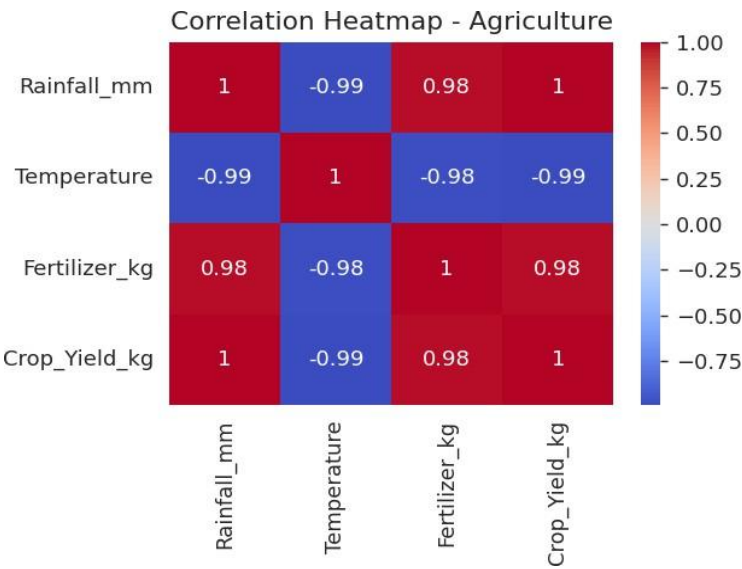


Figure 5.3

5.7 Trends, Distributions & Relationships

Rainfall and Crop Yield are almost perfectly correlated ($r = 0.996$), while Temperature is almost perfectly anti-correlated with both Rainfall ($r = -0.993$) and Yield ($r = -0.993$) - confirming a clear physical trend where cooler, wetter farms out-yield hotter, drier ones. The line chart (farms sorted by ascending rainfall) makes this co-movement visible directly: the yield line tracks the rainfall line almost step for step.

5.8 Key Findings & Insights

1. Rainfall shows a very strong positive correlation with crop yield ($r = 0.996$).
2. Loamy soil produces the highest average yield, followed by Clay, then Sandy.
3. Temperature is strongly inversely related to yield ($r = -0.993$); hotter farms ($32^{\circ}\text{C}+$) sit mostly on Sandy soil.
4. Fertilizer usage correlates positively with yield ($r = 0.983$), though slightly less strongly than rainfall.
5. Data cleaning found 0 missing values, 0 duplicates, and 0 IQR outliers.

5.9 Decision-Making Impact

Given rainfall's dominant effect, the department can prioritize irrigation investment for low-rainfall Sandy-soil farms first, and pair that with the fertilizer increases shown here to already be positively associated with yield